

VARUN MADASU

Java Full Stack Developer

Email: mvarun8031@gmail.com

Contact: +1 9135486558

linkedin: [linkedin.com/in/varun_madasu](https://www.linkedin.com/in/varun_madasu)



Professional Summary:

- Experienced Java Full Stack Developer with over 5 years of expertise designing and implementing scalable, high-performance web applications across diverse industries including enterprise support, retail, healthcare, and banking.
- Proficient in building microservices-based backend architectures using Spring Boot, Spring Cloud, and domain-driven design principles, ensuring maintainability and scalability.
- Skilled in modern frontend frameworks such as Angular and React, developing dynamic, responsive UI components with robust state management (NgRx, Redux) and seamless user experiences.
- Strong database design and optimization capabilities with PostgreSQL, MySQL, and Oracle, leveraging ORM frameworks like Hibernate and JPA for efficient data access and complex query handling.
- Expertise in securing applications using OAuth2, JWT, and role-based access control integrated with Spring Security and AWS IAM best practices.
- Extensive experience with containerization (Docker, Kubernetes/EKS), cloud platforms (AWS services including Lambda, S3, RDS), and Infrastructure as Code (Terraform, CloudFormation) to automate deployment and infrastructure management.
- Proven ability to build and maintain CI/CD pipelines using Jenkins, GitLab CI/CD, and AWS CodePipeline for continuous integration, testing, and deployment automation.
- Strong focus on software quality through comprehensive testing strategies utilizing JUnit, Mockito, Cypress, and Testcontainers, along with monitoring and logging using ELK Stack, Prometheus, and Grafana.
- Agile team player actively participating in Scrum and Kanban workflows, contributing to sprint planning, peer reviews, and mentoring junior developers to deliver business value efficiently and reliably.

Technical Skills:

Category	Skills & Technologies
Languages	Java 8/17, JavaScript, TypeScript
Backend	Spring Boot (2.x/3.x), Spring Cloud, Spring Security, Hibernate, JPA, Spring Batch
Frontend	Angular 14, React.js (Hooks), RxJS, NgRx, Redux Toolkit
UI & Styling	Angular Material, Tailwind CSS, HTML5, CSS3, SASS, Chart.js
Databases	PostgreSQL, MySQL, Oracle 11g
API & Auth	REST APIs, OAuth2, JWT, WebSockets (Spring STOMP)
Caching & Messaging	Redis, Apache Kafka
Cloud & Containers	AWS (RDS, Lambda, S3, DynamoDB, IAM), Kubernetes (EKS), Docker, Helm
CI/CD & DevOps	GitLab CI/CD, Jenkins, Maven, AWS CodePipeline, Terraform, CloudFormation
Testing	JUnit 5, Mockito, Testcontainers, Jest, Cypress
Monitoring	ELK Stack, Prometheus, Grafana
Methodologies	Agile Scrum, Kanban

Professional Experience:

HP, OK

Jan 2025 – Till Date

Java Full Stack Developer

Project: Enterprise Customer Support Portal

Overview: The Enterprise Customer Support Portal is a scalable web application designed to manage customer service operations, including ticketing, live chat, SLA tracking, and performance analytics. The platform supports enterprise clients by integrating backend microservices with a modern Angular frontend to deliver responsive and secure user experiences for both customers and support agents.

Responsibilities:

- Designed and implemented RESTful microservices using Spring Boot, Spring Cloud, and Java 17 to handle ticket lifecycle management, authentication, SLA tracking, and user management. Applied Domain-Driven Design (DDD) for scalable service boundaries.
- Built dynamic, modular frontend components using Angular, RxJS, and NgRx for state management. Developed role-based dashboards and interactive charts using Chart.js and Angular Material.
- Modeled and optimized relational schemas using PostgreSQL and developed complex SQL queries for reporting modules. Used JPA/Hibernate for ORM, ensuring high performance with batch processing and lazy/eager loading strategies.
- Integrated Spring Security with OAuth2 and JWT-based token management for secure, role-based access control across the application.
- Configured Netflix Zuul and Spring Cloud Gateway for API routing and load balancing. Managed service discovery using Eureka and centralized configuration with Spring Cloud Config Server.
- Worked closely with DevOps to configure GitLab CI/CD pipelines. Wrote Dockerfiles for each microservice, and deployed containers to Kubernetes (EKS) environments using Helm charts and kubectl.
- Developed comprehensive unit and integration tests using JUnit 5, Mockito, and Testcontainers. Contributed to frontend test coverage using Karma, Jasmine, and Cypress for E2E flows.
- Improved performance using Redis for caching frequently accessed ticket data. Integrated Apache Kafka for asynchronous event-driven workflows like ticket status updates and email triggers.
- Implemented centralized logging using ELK Stack (Elasticsearch, Logstash, Kibana) and application performance monitoring with Prometheus and Grafana.
- Participated in Scrum ceremonies, peer code reviews, and sprint planning. Used Jira for backlog grooming and story tracking, and Confluence for documenting technical decisions.

Environment: Java 17, Spring Boot, Angular 14, PostgreSQL, Redis, Kafka, Docker, Kubernetes, GitLab CI/CD, ELK Stack, Prometheus.

Albertson, CA

April 2024 – Dec 2024

Java Full stack Developer

Project: Real-Time Stock Management Platform

Overview: Developed a scalable web application to enable real-time inventory synchronization across multiple retail outlets, improving stock accuracy, operational efficiency, and visibility through automated updates and centralized control.

Responsibilities:

- Designed responsive and modular UI components using React.js (Hooks, functional components) with Tailwind CSS for consistent styling.
- Managed global application state and complex UI interactions with Redux Toolkit, utilizing Redux Thunk for async actions related to API calls (e.g., inventory sync, store switch).
- Applied React Router for route-based navigation and lazy loading to optimize performance.
- Integrated form validations using React Hook Form and Yup, and implemented client-side caching for better UX.
- Conducted unit testing with Jest and React Testing Library to ensure reliability of frontend logic.
- Built RESTful APIs in Spring Boot 3.x with layered architecture (Controller, Service, DAO) for clear separation of concerns.
- Implemented business logic to handle stock transfers, updates, threshold triggers, and SKU versioning.
- Used Spring Security for JWT-based authentication and authorization, with role-based access control (Admin, Store Manager, Viewer).
- Integrated Hibernate 6 for ORM, creating entity relationships (OneToMany, ManyToOne, @JoinTable) and using JPQL and Criteria API for dynamic queries.
- Created batch processing modules with Spring Batch for periodic data reconciliation and alert generation.
- Designed normalized schemas with referential integrity for product inventory, store mappings, and transaction history.
- Tuned SQL queries and used indexes, materialized views, and partitioning for performance optimization.
- Created stored procedures and triggers for automating stock adjustment on API updates.
- Leveraged pgAdmin and Flyway for database versioning, migration, and backups.
- Designed and exposed REST APIs using Spring MVC, with JSON payloads adhering to OpenAPI standards.
- Integrated third-party services (e.g., vendor APIs, alert notification system) using WebClient (Reactive HTTP client).
- Implemented real-time inventory broadcasting with WebSockets using Spring's STOMP support.
- Enabled Swagger UI for auto-generated API documentation to support frontend/backend collaboration and QA.
- Used Docker Compose to containerize and manage isolated development environments (backend, frontend, database).
- Automated build and deploy pipelines using Jenkins and Maven, including test phases and static code analysis.
- Maintained version control via Git (GitHub), managed branching strategy, and peer-reviewed merge requests.
- Worked in Agile Scrum environment, using JIRA for sprint tracking, story point estimation, and defect resolution.

Environment: React.js, Redux Toolkit, Spring Boot 3, Hibernate 6, PostgreSQL 14, REST APIs, Docker, Git, Jenkins, Agile Scrum.

State Of Louisiana, LA

Sep 2021 – Nov 2023

Software Engineer

Project: HealthSync Platform

Overview: The HealthSync Platform is a secure, full-stack web application developed to streamline and manage healthcare-related Data Update Requests (DURs) while ensuring strict compliance with HIPAA regulations.

Responsibilities:

- Designed and developed scalable full-stack web applications by building responsive React.js frontends and robust Java Spring Boot backends, leveraging JDBC, MySQL, and advanced SQL optimization techniques to support complex Data Update Requests (DURs) and ensure high-performance data handling.
- Designed and developed RESTful APIs using Spring Boot, facilitating seamless integration between frontend interfaces and backend services.
- Utilized Hibernate and JPA for efficient data persistence and ORM, ensuring data integrity, optimizing SQL query performance, and supporting scalable transactional operations.
- Integrated AWS RDS for relational data storage and leveraged AWS-managed services for enhanced scalability, reliability, and fault tolerance.
- Developed responsive, accessible user interfaces using React.js, HTML5, CSS3, and SASS, collaborating with UX/UI designers to transform wireframes into functional, production-ready components.
- Implemented state management and data binding using React hooks, Context API, and RxJS observables, improving frontend responsiveness and user experience.
- Leveraged AWS Lambda, S3, and DynamoDB to build serverless, event-driven architectures, enabling cost-effective, scalable backend solutions.
- Configured CI/CD pipelines using Jenkins and AWS CodePipeline to automate build, test, and deployment workflows, ensuring continuous integration and delivery.
- Defined Infrastructure as Code (IaC) using AWS CloudFormation and Terraform, supporting consistent and reproducible cloud infrastructure provisioning.
- Implemented secure authentication using OAuth2 and JWT token-based authorization, and enforced SSL/TLS encryption for secure data transmission.
- Managed user roles and least privilege access across cloud resources using AWS IAM roles and policies, maintaining security best practices and compliance.
- Led Root Cause Analysis (RCA) for production issues using Eclipse IDE, implemented unit testing with JUnit, ensured HIPAA compliance in healthcare data processing, and contributed to Agile sprint deliveries by mentoring junior developers, conducting QA reviews, and supporting CI/CD pipelines with Git.

Environment: Java, JavaScript, TypeScript, Angular, HTML5, CSS3, SASS, Spring Boot, Hibernate, JPA, MySQL, AWS RDS (Relational Database Service), AWS, Docker, Git, GitHub, Jenkins, AWS Code Pipeline, JUnit, Mockito, Selenium, RESTful APIs, OAuth2, JWT, AWS IAM (Identity and Access Management), SSL/TLS, Scrum, Kanban.

BB&T, India

July 2019 – Aug 2021

Java/J2EE Developer

Project: Online Banking Alerts Module

Overview: Designed and developed the Online Banking Alerts Module, enabling real-time notifications for customer account activities, security events, and transactional updates.

Responsibilities:

- Followed Agile (Scrum) methodology to meet customer expectations and deliver high-quality solutions on time.
- Designed and developed user interaction screens using HTML5, CSS3, JavaScript, jQuery, and AJAX.
- Developed backend modules in Java and integrated with MySQL using JDBC.
- Participated in code maintenance, debugging, and enhancement activities.
- Built web-based applications using JavaServer Faces (JSF), Spring MVC, and Spring Web Flow.
- Managed Java JDK setup, including classpath and JVM configuration.
- Developed client-side scripts with JavaScript and implemented RESTful services using JSON.
- Utilized Spring annotations and XML for Dependency Injection and application configuration.
- Created Spring JDBC connections to Oracle 11g and wrote SQL queries for data manipulation.
- Developed Hibernate configuration files, mapping files, and entity classes for persistence operations.
- Represented the team in project meetings and client interactions.
- Prepared and executed test plans using JUnit to ensure application quality.
- Designed UML artifacts including use cases, sequence diagrams, and behavioral diagrams for the Alerts module.
- Applied AJAX for dynamic user operations and front-end validation.

Environment: Java SE 8, Soap UI, Junit, Agile, JDK 1.7, J2EE 1.5, Spring 2.0, Hibernate, Eclipse Juno.

Education:

Masters: University of Central Missouri - Missouri

Bachelors: Vasavi College of Engineering - INDIA

Certifications:

- Certified AWS Developer